

Scenario 1: Using historical forecast data to evaluate past flood events

Description

As a floodplain manager for a mid-sized municipality, you are working on updating your city's flood hazard mitigation plan. Your team wants to better understand the frequency and magnitude of flood events from the past 8-10 years.

Your Task

- Select a region of interest from the HOME page or locate the region on the MAPS page.
- Click the icon with three stacked layers at the top right of the map to open the map layers menu. Select your preferred basemap.
 - **ESRI World Imagery** displays high-resolution satellite imagery, which is useful for visually identifying landmarks and natural features.
 - **USGS Imagery** offers detailed topographic information that helps in interpreting terrain in flood-prone areas.
 - **CartoDB Positron** provides a cleaner, less detailed background, ideal for users who prefer a minimal map to focus solely on the flood extent.

**Note: For best performance, we recommend using “ESRI World Imagery” as the basemap. As of August 2025, selecting “USGS Imagery” may cause flood maps to not display properly.*

- Zoom in to find a river location of interest (represented by blue lines on the map). Click on a river to select it. You will see some information pop up including the name, river ID, etc., of the selected river.
- Use the **Historical Data** feature (button found at the top left of the page) to visualize the historical streamflow time-series for a location of interest. You will see a graph pop up showing streamflow (in cubic feet per second) over time (by default up to 90 days in the past).
- To display historical streamflow data for a specific event in the location of interest, click on the date selector icon on the top right of the graph. Select the desired date range. **Note: Historical near-real-time streamflow data has been available since 2016, the year the National Water Model (NWM) became operational.*
- If flood maps are available for the selected river, a vertical slider will appear on the right side of the page. This slider allows you to explore flood scenarios under different streamflow conditions. As you move the slider, the corresponding pre-computed flood extents will display on the map, indicating areas of possible inundation. **Note: These maps are pre-generated for a set of representative conditions, so the one displayed may not match the exact streamflow or stage value you are interested in, but it will reflect the closest available condition.*
- Cross-reference this information with flood stage threshold (minor/moderate/severe) data available on the NWM. The NWPS/NWM tutorial below provides steps on how to access this information.
- Evaluate whether this tool provides sufficient detail to identify vulnerable areas and support future planning decisions around flood mitigation.